

6 Related Work

Long-context Language Models The context window of language models has significantly increased due to recent efforts in training long-context language models with optimized data mixture (Grattafiori et al., 2024; Gao et al., 2025; AI et al., 2025), exploring novel model architectures (Gu and Dao, 2024; Dao and Gu, 2024; Fu et al., 2023; Peng et al., 2023; Bertsch et al., 2023), developing encoding mechanisms (Su et al., 2023; Yen et al., 2024) as well as length extrapolation techniques (Press et al., 2022; Sun et al., 2022; Chen et al., 2023).

Long-context Benchmarks Many existing benchmarks evaluate language models’ long-context abilities along multiple dimensions. ZeroSCROLLS (Shaham et al., 2023) focus on long-context information gathering tasks. Long-bench (Bai et al., 2024) introduces a bilingual benchmark and studies context-length variations. L-Eval (An et al., 2023) aim for a more diverse and high-quality collection of datasets. Infinite-Bench (Zhang et al., 2024) extend the context-length to over 100K tokens. HELMET (Yen et al., 2025) presents a holistic evaluation beyond synthetic data. Other previous work focus on specific tasks, such as retrieval-augmented generation (Lee et al., 2024), question answering (Kočíský et al., 2017; Wang et al., 2025), in-context learning (Agarwal et al., 2024; Anil et al., 2024; Xu et al., 2024), coreference resolution (Vodrahalli et al., 2024), and document summarization (Chang et al., 2024; Kim et al., 2024). Compared to those canonical tasks designed for benchmarking long-context understanding abilities, *AbsenceBench* features the task of handling long contexts at a finer granularity. Similar to the NIAH test (Kamradt, 2023; Yen et al., 2025) that evaluates the ability of LLMs to locate and retrieve crucial pieces of information, we formulate *AbsenceBench* into a much more straightforward task of identifying omissions in context, which turns out to be more challenging than identify the presence of information.

7 Limitations

While *AbsenceBench* exposes fundamental challenges in how current LLMs process omitted content, we note several limitations.

Surface-form omission only. We chose to keep the tasks exceedingly simple; even though our benchmark is solvable by simple heuristics models fail to use these heuristics. Our benchmark exclusively evaluates surface-form omission, where the removed content is a simple deletion. This simplification was intentional and helps make the evaluation unambiguous and trivially automatic. However, it does not capture more subtle or semantic notions of absence—such as omitted reasoning steps or missing evidence—which are critical in many real-world applications like rubric-based grading or code review.

Structured task format. *AbsenceBench* presents models with both the original and modified documents side-by-side and asks for the omitted elements directly. This setup may not reflect naturalistic settings where users do not provide such explicit comparisons. As a result, our evaluation may overestimate how well models would perform under more realistic situations with omitted information. Still, given the poor performance on *AbsenceBench*, current models are clearly not performant. However, if a new generation of models solve *AbsenceBench*, that will not be sufficient evidence to say that models can handle omission elegantly.

Evaluation scope. Our benchmark covers three domains (poetry, numerical sequences, and GitHub pull requests) with medium-length contexts (5K tokens). While this breadth offers early insight into the difficulty of absence detection, our findings may not generalize to other modalities (e.g., vision, audio), domains (e.g., legal, medical), or longer contexts. However, we expect these different settings to be more difficult, not less.

No prompt or instruction tuning. We only explored two different prompts and did not explore prompt tuning or in-context examples, due to the overwhelming cost of running long-context models, especially the most performant ones that make use of inference-time compute. Note that many inference-time models consistently use more the three times more “thinking tokens” than the original length of the document they are scanning for omissions. Thus, it remains an open question whether

more elaborate prompting strategies, could significantly improve model performance, something we hope to examine in future work.

Statistical significance. Finally, we did not compute error bars or significance testing for our evaluation across runs due to API cost constraints. Although our findings are consistent across tasks and models, additional replication would help quantify variability and confirm robustness. We especially would have liked to run a document perturbation study, to see whether small variations in the document or prompt affect performance.

8 Conclusion

We introduce `AbsenceBench`, a benchmark that tests LLMs’ ability to detect omitted information—an ability distinct from the well-studied task of recalling present content. Despite the benchmark’s simplicity, models perform poorly, with surprising trends: fewer omissions make the task harder, and inference-time compute models often generate three times as many “thinking tokens” than the document length itself. Explicitly inserting placeholders where content is missing substantially improves performance, supporting the hypothesis that attention struggles to represent absence. These results reveal a core limitation in current models and motivate future work on absence-aware architectures and evaluation.

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